

Midterm Exam 1 Formula Sheet (Chapters 1-5)

Chapter 3

Displacement: $\Delta x = x_f - x_0$ Average velocity: $\bar{v} = \frac{x_f - x_0}{t_f - t_0}$ Instantaneous velocity: $v = \frac{dx}{dt}$ Average acceleration: $\bar{a} = \frac{v_f - v_0}{t_f - t_0}$ Instantaneous acceleration: $a = \frac{dv}{dt} = \frac{d^2x}{dt^2}$

Kinematic equations:

$$\bar{v} = \frac{v + v_0}{2}$$

$$v = v_0 + at$$

$$x = x_0 + v_0t + \frac{1}{2}at^2$$

$$v^2 = v_0^2 + 2a(x - x_0)$$

Acceleration due to gravity:

$$a = -g = -9.80 \text{ m/s}^2$$

Chapter 4

Kinematic equations for projectile motion:

$$x = x_0 + v_{0x}t$$

$$v_y = v_{0y} - gt$$

$$y = y_0 + v_{0y}t - \frac{1}{2}gt^2$$

$$v_y^2 = v_{0y}^2 - 2g(y - y_0)$$

$$\text{Maximum height: } h = \frac{v_0^2 \sin^2 \theta_0}{2g}$$

$$\text{Horizontal range: } R = \frac{v_0^2 \sin 2\theta_0}{g}$$

$$\text{Total time of flight: } t = \frac{2v_0 \sin \theta_0}{g}$$

Uniform circular motion:

$$a_c = \frac{v^2}{r}$$

$$T = \frac{2\pi r}{v}$$

Chapter 5

Newton's second law of motion: $\vec{F}_{net} = m\vec{a}$

Component form of Newton's second law:

$$\Sigma F_x = ma_x$$

$$\Sigma F_y = ma_y$$

$$\Sigma F_z = ma_z$$

Static friction: $f_s \leq \mu_s N$ Kinetic friction: $f_k = \mu_k N$ Where μ_s are μ_k coefficients of static and kinetic friction, respective, N is normal forceSpring force: $F = -k\Delta x$

where k is the spring constant

Extra equations

Trigonometry:

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$$

$$\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}$$

$$\tan \theta = \frac{\text{opposite}}{\text{adjacent}}$$

Solutions to the quadratic equation

$$ax^2 + bx + c = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$